

Week 05 Task 13

Phishing Email Analysis

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Table of Contents

1) Executive Summary 3
2) Sample Email 3
3) Read Headers, SPF, DKIM, Return Path, From 4
4) Check a suspicious url on virus total 5
5) List the Red flags 6
6) Verdict 6
7) TryHackMe Room Phishing Emails 7
8) Conclusion 7

1) Executive Summary

This task focused on phishing email analysis, including examining a suspicious email, analyzing its headers and authentication protocols, and identifying potential phishing indicators. The investigation also involved checking the suspicious URL using VirusTotal to assess its reputation and reviewing the email for common phishing characteristics. Additionally, the TryHackMe room Phishing Emails in Action was completed to gain practical experience in identifying and analyzing phishing emails.

2) Sample Email

From: "noreply@Quick Response" nuthostsrl.SaintU74045Walker@comunidadeduar.com.ar
To: Phishingisfun@hotmail.com
Date: Friday, 08 September 2023, 10:11 AM (UTC+5)
Subject: We locked your account for security reason - Fri, September 08, 2023 10:11 AM
Return-Path: nuthostsrl.SaintU74045Walker@comunidadeduar.com.ar
Attachment: Detailsdisable-262340.pdf

Hello Phishingisfun@hotmail.com**,**

Your Amazon account has been put on hold

We took this action because the billing information you provided did not match the information in the card issuer's data.

Please update your information as soon as possible so you can continue using your Amazon card.

[Update Information]

https://script.google.com/macros/s/AKfycbwmwl6-oDDgtuO7lmFuHnvuNd-mvDQlzNJaRxbTkdag0Q7lygpC3YzQqCKpTWl-aWsCqw/exec

In order to maintain the safety of your account, your account will be on hold until you fulfill the required forms.

You might want to do this sooner; any locked account will be deleted in order to protect the data from being leaked.

Sincerely,
Amazon Team Support

You received this email to let you know about important changes to your Amazon account and services.

1996–2023, Amazon.com, Inc. or its affiliates.

3) Read Headers, SPF, DKIM, Return Path, From

The email headers were examined to identify the sender's information, return path, sending domain and email authentication results.

Header Field	Extracted Value
From	noreply@Quick Response nuthostsrl.SaintU74045Walker@comunidadeduar.com.ar
Return-Path	nuthostsrl.SaintU74045Walker@comunidadeduar.com.ar
To	Phishingisfun@hotmail.com
Subject	We locked your account for security reason
Date	08 September 2023
Sender IP	40.107.215.98

Email Authentication Results

SPF

The SPF authentication check passed. This indicates that the sending IP address was authorized by the SPF policy of the domain used for the sender's email address. However, SPF authentication does not verify whether the sender is genuinely associated with Amazon.

DKIM

The DKIM authentication result was none, indicating that the email was not signed with a DKIM signature. Therefore, no DKIM signature was available for the receiving server to verify the message's integrity and signing domain.

DMARC

The DMARC authentication result was bestguesspass, rather than a standard pass. This indicates that the receiving server returned a best-guess authentication result instead of confirming a normal DMARC pass. It should not be interpreted as definitive proof of successful DMARC authentication.

Header Analysis

The email uses the display name **"noreply@Quick Response"** and claims to be an Amazon account security notification. However, the actual sender address belongs to **comunidadeduar.com.ar**, which is unrelated to Amazon.

The Return-Path matches the From email address, so there is no mismatch between these two fields. Nevertheless, the sender domain raises suspicion because it does not match Amazon's official domain.

Email authentication analysis showed that **SPF passed**, **DKIM was absent (dkim=none)**, and **DMARC returned bestguesspass**, rather than a standard pass result. These results do not establish that the email originated from Amazon or that the message is legitimate.

Finding: The email contains suspicious sender information and impersonates Amazon using an unrelated sender domain. The authentication results do not verify the claimed Amazon identity.

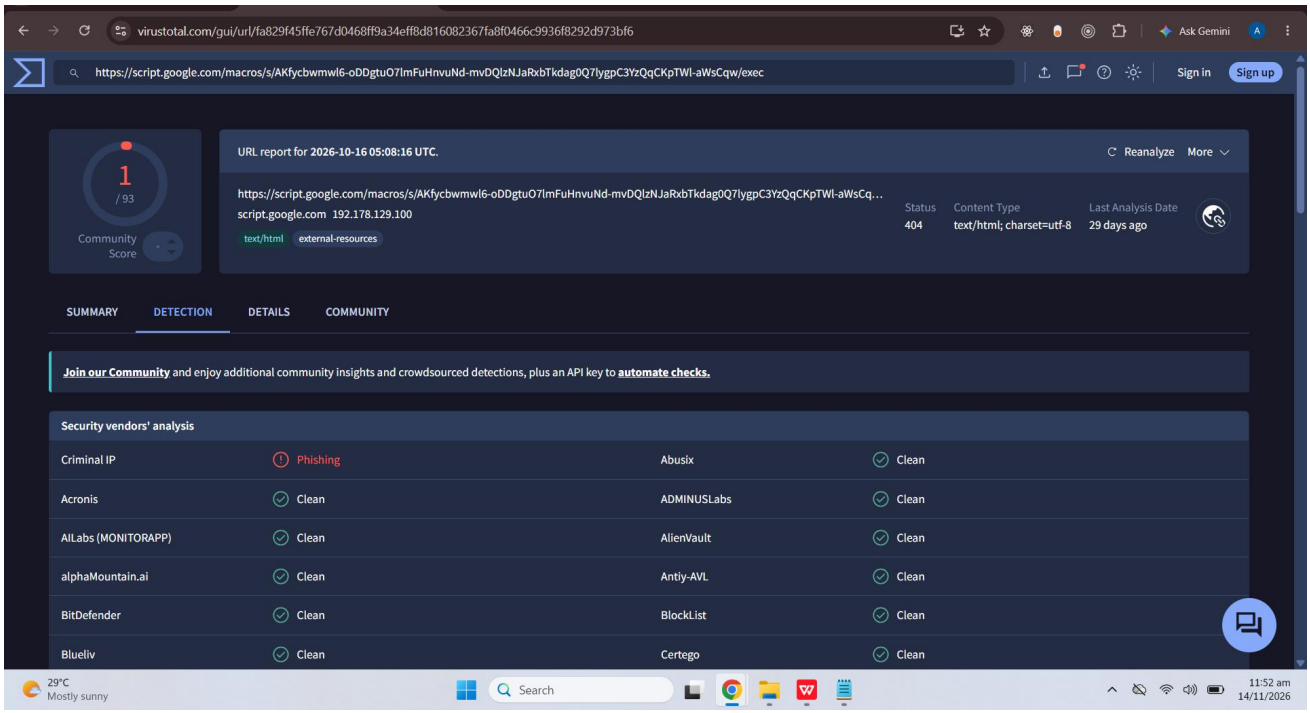
4) Check a suspicious url on VirusTotal

Objective

To investigate the suspicious URL extracted from the phishing email and determine whether it has been identified as malicious or associated with phishing activity.

Suspicious URL

https://script.google.com/macros/s/AKfycbwmwl6-oDDgtuO7lmFuHnvuNd-mvDQlzNJaRxbTkdag0Q7lygpC3YzQqCKpTWl-aWsCqw/exec



Virus Total Results

Indicator	Result
Detection Ratio	1/93
Flagging vendor	Criminal IP
Detection classification	Phishing
Other displayed vendors	Clean
HTTP Response Status	404
Content-Type	text/html; charset=utf-8

VirusTotal reported that 1 out of 93 security vendors flagged the submitted URL as phishing. Criminal IP identified the URL as phishing, while the other displayed vendors classified it as clean.

The URL uses the legitimate Google Apps Script domain (script.google.com). However, legitimate cloud services can be abused by attackers to host phishing content.

The URL also returned an HTTP 404 (Not Found) status in the recorded scan, indicating that the requested resource was unavailable at that time.

Although the detection ratio is relatively low, the phishing classification from Criminal IP, combined with the suspicious email content and the use of an unrelated sender domain impersonating Amazon, raises security concerns.

5) List the Red flags

1. **Brand Impersonation:** The email claims to be from Amazon but uses Amazon branding without originating from an official Amazon domain.
2. **Suspicious Sender Domain:** The sender uses **comunidadeduar.com.ar**, an unrelated domain rather than an official Amazon domain.
3. **Urgent and Threatening Language:** The email threatens account deletion to pressure the recipient into taking immediate action.
4. **Suspicious URL:** The "Update Information" button redirects to a Google Apps Script URL instead of an official Amazon website.
5. **Attachment File-Type Mismatch:** The attachment **Detailsdisable-262340.pdf** contains HTML markup instead of genuine PDF content.
6. **Missing DKIM Signature:** The authentication results show **dkim=none**, meaning the email was not signed with a DKIM signature.
7. **VirusTotal Detection:** Criminal IP flagged the suspicious URL as phishing, with 1 out of 93 security vendors detecting it.

6) Verdict

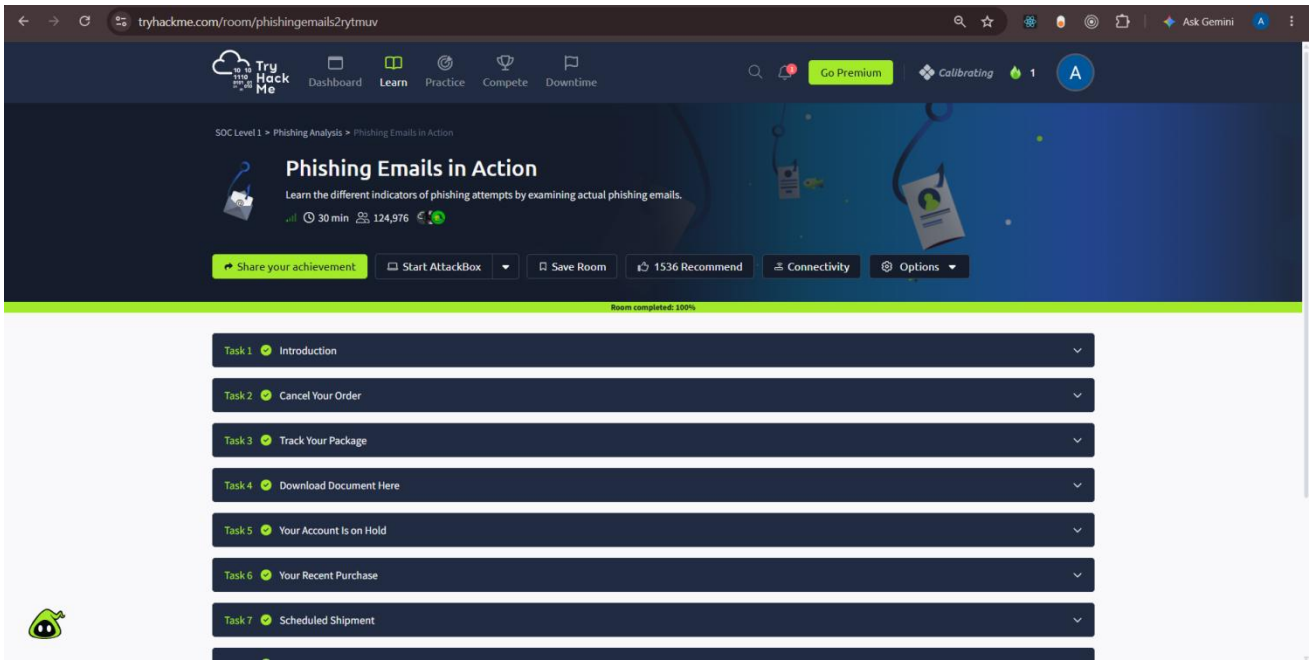
Based on the investigation and analysis of the email headers, sender information, embedded URL and attachment, the email has been identified as a **phishing attempt impersonating Amazon**.

The email uses Amazon's branding and claims that the recipient's account has been placed on hold due to billing issues. It creates urgency by threatening account deletion and encourages the recipient to update their information through a suspicious Google Apps Script URL rather than an official Amazon website.

The sender address belongs to **comunidadeduar.com.ar**, which is unrelated to Amazon. Although SPF authentication passed, the email had no DKIM signature, and DMARC returned a **bestguesspass** result rather than a standard pass.

Furthermore, the attached file, **Detailsdisable-262340.pdf**, contains HTML markup instead of genuine PDF content, indicating a file-type mismatch. VirusTotal analysis also identified the embedded URL as phishing through one security vendor (Criminal IP), out of 93 vendors.

7) TryHackMe Room Phishing Emails



As show in the above figure I have Completed the TryHackMe room by examining different phishing email scenarios, including fake order cancellations, package tracking notifications, document download requests, account-hold alerts, recent purchase notifications, and scheduled shipment messages. Practiced identifying phishing indicators and recognizing common email-based social engineering techniques. The room was completed with 100% progress

8) Conclusion

The phishing email investigation successfully demonstrated how SOC analysts can identify and assess suspicious email activity through structured analysis and threat intelligence tools. The investigation identified several phishing indicators, including Amazon brand impersonation, an unrelated sender domain, urgent and threatening language, a suspicious Google Apps Script URL, and an attachment containing HTML markup despite having a PDF extension. The email authentication results also demonstrated that SPF passing does not guarantee an email's legitimacy. DKIM was absent, while DMARC returned a best-guess result. Furthermore, VirusTotal identified the submitted URL as phishing through Criminal IP, although the overall detection ratio was only 1 out of 93 vendors. The findings support classifying the email as a phishing attempt. The investigation reinforced the importance of correlating multiple indicators rather than relying on a single security tool or authentication result. As a practical learning activity, the TryHackMe room Phishing Emails in Action was also completed to strengthen understanding of phishing email analysis and common social engineering techniques.